Land-Use Classification of High-Resolution Remote Sensing Imagery Incorporating Global–Local Interactive Features Across Optical Domain and Land Cover Primitives

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***Abstract*—The land use (LU) classification of remote sensing (RS) images has broad applications in various fields. In recent years, hybrid convolutional neural networks (CNNs)–Transformer models have been widely applied to the LU classification task. These methods combine the advantages of CNNs and transformers to mine information from RS images fully for LU classification. However, due to rich content and complex backgrounds in RS images, existing methods often overlook the deep semantic information and spatial structural relationships hidden within the optical domain, resulting in limited classification accuracy. To address these challenges, we propose a novel LU classification model: the global–local interactive feature fusion model (GLIFFM). The GLIFFM utilizes a CNN–Transformer architecture and simultaneously integrates land cover (LC) primitives to fully exploit both semantic content within the RS images and the spatial structural information. First, a mutual feature extraction (MFE) module is designed to extract correlated and complementary features between optical and LC primitives. Next, we introduce a local and global feature enhancement module (LGFEM) to enhance deep features containing both LC and optical information. Finally, a gated information fusion block (GIFB) is proposed to perform gated fusion of the local and global features. The experimental results conducted on the WHDLD, EuroSAT, and LCZ-Guangdong-Hong Kong-Macao Greater Bay Area (LCZGBA) datasets demonstrate that our GLIFFM can achieve superior classification performance and outperform many state-of-the-art methods. Our source codes are available at https://github.com/Fimwu7/The-GLIFFM-for-Land-Use-Classifcation.**

***Index Terms*—Feature fusion, land cover (LC), land-use (LU) classification, remote sensing (RS).**

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I. Introduction

LAND-USE (LU) mapping is important to provide a data basis for decision-making across various fields, including Earth resource management, environmental protection, and urban and rural planning and development [1]. Unlike land cover (LC) (e.g., forest and water), which describes the physical characteristics of the Earth's surface, LU (e.g., residential land and parks) is determined by human socio-economic activities in a land. This distinction presents a fundamental challenge for LU classification using remote sensing (RS) imagery: satellite or aerial imagery inherently captures the spectral and textural attributes of LC but lacks direct information about socio-economic function [2]. Consequently, LU classification based solely on RS imagery faces inherent difficulties characterized by high intraclass diversity and high interclass similarity [3].

To bridge this semantic gap, researchers increasingly leverage multisource geospatial data for LU classification alongside RS imagery [4], e.g., nighttime light data [5], open street map [6], crowdsourced data [7], [8], geospatial big data [9], point-of-interest data [10], [11], [12], and voluntary geographic information [13], [14]. These datasets provide vital socio-economic context and functional semantics, thus demonstrably improving LU classification through complementary information dimensions.

Nevertheless, integrating these multisource data presents challenges. The structural and spatiotemporal mismatches between RS images and auxiliary data make it difficult to establish deep correlations between modalities. Furthermore, auxiliary sources like voluntary geographic information can suffer from issues like outdated information, incompleteness, and inaccuracy, which further impact model accuracy and generalization. Effective strategies are still needed to address these concerns.

As for LU classification using only RS images, deep learning models, particularly convolutional neural networks (CNNs) and Transformer models, have recently become mainstream tools for extracting deep visual features. A significant number of methods based on both CNN [15], [16], [17], [18] and Transformers [19], [20], [21] have been proposed for LU classification, achieving remarkable results. CNNs focus on extracting features from local regions. Benefiting

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from their unique hierarchical structure, many researchers have achieved excellent performance in LU classification by employing methods that hierarchically fuse local features extracted by CNNs [22], [23], [24]. However, CNNs are hindered by their inherent inductive biases, which prevent them from modeling long-range dependencies. Therefore, some researchers have attempted to augment CNNs' feature representation capability by incorporating multiscale contextual information [25], [26]. Nevertheless, these approaches fundamentally fail to resolve CNNs' inherent limitations in modeling global, nonlocal dependencies. Even with multiscale contextual information, CNNs' inherently limited receptive fields and sequential processing nature still hinder their ability to efficiently and comprehensively capture complex, long-range global relationships within an image, particularly when dealing with truly distant pixel correlations.

With the development in computer vision, Transformers have increasingly shown their promise in LU classification tasks. Their core self-attention mechanism enables the extraction of dependencies between pixels at arbitrary positions within an image, thereby directly modeling long-range dependencies and global contextual information. This inherent global perception capability allows Transformers to overcome the limitations of CNNs in processing complex RS imagery. Extensive research has demonstrated that Transformers' ability to capture global features provides significant advantages for LU classification [27], [28]. However, compared to CNNs, Transformers typically possess a large number of parameters and exhibit slower convergence, necessitating more computational resources and time. In addition, the Transformers' self-attention mechanism, which learns relationships between patches, prevents it from efficiently capturing hierarchical local features. Consequently, the integration of CNNs and Transformers has emerged as a mainstream research direction due to their complementarity. For example, Zheng et al. [29] proposed LDBST, a network featuring two branches: a Vision Transformer branch for global information and a CNN branch for local information. In [30], a sequential CNN–Transformer was proposed for RS scene classification. Specifically, stacked CNN modules were employed to extract multilevel local features, and these features were fed into a LightFormer block to capture long-range dependencies. These CNN–Transformer methods effectively combine the advantages of CNNs and Transformers, achieving significant results in LU classification tasks [31], [32].

These deep learning methods, whether based on CNNs, Transformers, or CNN–Transformer, excel at efficiently extracting fine visual features (like building shapes, road textures, or vegetation density) from optical images. However, their fundamental nature involves mapping these visually perceptible features to predefined high-level semantic LU categories. This mapping mechanism proves inadequate for the concept of LU, as the model's perception is limited to what image pixels can express. The true meaning of LU often transcends purely visual categories, encompassing a significant amount of nonvisual, socio-economic information.

LC and LU are two ways of observing Earth's surface, answering two basic questions: What is this (LC)? What is it for (LU) [33]? LU is a cultural concept describing human activities and their use of land, whereas LC is a physical description of the land surface [34]. Essentially, LU is a higher-level, socio-semantic expression of LC. While LC can be used to infer LU, the two concepts are not entirely interchangeable [35]. More accurately, LU represents the systematic, non-random spatial organization, configuration, and restructuring of various LC types within a specific geographical spatial unit. This organization is driven by human socio-economic objectives and needs, ultimately embodying or reflecting its mapped specific functions and activities. However, some identical low-level ground features or LC classes are frequently shared amongst different LU categories [36]. Some existing studies have confirmed the presence of multilevel feature relationships between LU and LC. For instance, research by [37] and [38] explored their spatial nestedness and hierarchical relationships. While these studies consistently demonstrate performance improvements in LU classification, there remains a notable scarcity of research conducting deeper, interactive learning between LU and LC. To fill this research gap, this study aims to explore and propose a novel method to achieve more tightly coupled deep interactive learning between LU and LC, thereby further enhancing the accuracy and semantic understanding of LU classification.

To address the aforementioned challenges, i.e., the semantic gap in LU classification, the limitations of auxiliary data integration, and the need for robust feature fusion across optical and LC primitives, we propose a novel global–local interactive feature fusion model (GLIFFM). This model uniquely leverages LC information derived directly from the input RS imagery itself as a complementary semantic resource, bypassing external data dependency and its associated problems. More importantly, it tries to exploit global–local mutual features across optical and LC primitives, offering novel support for the traditional LU classification tasks using multimodal data. Specifically, we introduce an interactive information embedding module that captures intrarepresentation interaction across optical and LC primitives in addition to their inner-representation interaction. Subsequently, we design a mutual feature extraction (MFE) module to achieve deep, cross-asymmetric fusion of optical visual features and high-level semantic features mapped by the derived LC primitives. This multipath structure is specifically engineered to uncover and leverage the complementary clues embedded within the shallow features of these two intrinsically linked modalities. By explicitly modeling the inner/intrarepresentation in/across optical and LC primitives, the MFE effectively mitigates the high intraclass diversity and high interclass similarity problem in LU classification with only visual features. Simultaneously, we introduce a plug-and-play global–local feature enhancement and fusion modules to integrate the capabilities of both Transformer and CNN. It prioritizes the interaction of local and global spatial features using a gated fusion mechanism within a multilevel branching structure. This introduces beneficial inductive bias through grouped fusion, adaptively weighting the contribution of features at different scales and fostering a significantly stronger, more generalizable feature representation that synergizes CNN-like local processing with

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Transformer-like global context modeling. The main contributions of this article are summarized as follows.

1) We propose GLIFFM, a CNN–Transformer hybrid network specifically designed to integrate implicitly derived LC information with optical RS imagery for LU classification. By considering the varying contributions of global–local interactive features across optical and LC primitives, our method effectively uncovers complementary information within LC data.
2) This article introduces an MFE module, enabling effective multipath, cross-asymmetric fusion of optical and LC-derived semantic features. This mechanism is explicitly designed to learn deep and complementary insights across different modalities by linking physical form (LC) to socio-economic function (LU).
3) We developed a plug-and-play dynamic global–local feature enhancement and fusion module. Its multigrouped and gated fusion strategy injects inductive bias, enhancing the model's ability to learn robust and discriminative feature representation.
4) We construct a specialized LU classification dataset on the basis of the WHDLD dataset with pixel-level LC labels [39], [40] by visually determining LU type for each image patch.

The remainder of this article is organized as follows. Section II reviews related work on LU classification methods. Section III provides a detailed description of the proposed GLIFFM. Section IV presents comprehensive experiments, ablation studies, and results analysis across three datasets. Section V provides a discussion on the model's mechanisms and limitations. Finally, Section VI concludes the article and suggests future research directions.

II. Related Work

A. LU Classification Based on CNN

Due to the hierarchical structure, CNNs can automatically learn rich semantic information from RS images. Consequently, CNN-based methods have been widely applied to LU classification tasks. Nevertheless, isolated local features might not adequately capture LU information. To address the inherent limitations of CNNs in LU classification, researchers have proposed diverse solutions. For example, Chen et al. [41] proposed a contextual information-preserved architecture learning (CIPAL) framework, by channel compression, to utilize the contextual information in RS images as much as possible during the architecture learning process. By incorporating contextual information into CNNs, this approach can achieve promising results in LU classification tasks. With the aim of strengthening the model's discriminative performance and its ability to capture rich information. Li et al. [42] utilized multilevel attention to design a gated recurrent multiattention neural network (GRMA-Net). This attention mechanism allows the network to focus on information at different levels, thereby extracting spatial features that are then fed into a gated recurrent unit (GRU) to capture contextual relationships. Experimental results demonstrate that GRMA-Net achieves excellent classification performance on multiple RS image datasets. An efficient systematic architecture, named multiscale feature fusion network with enhanced feature correlation (EFCOMFF-Net), proposed in [43], aims to reduce the gap among multiscale features and fuse them to improve the representation ability of RS images. Furthermore, numerous other works ([44], [45], [46], [47]) integrate additional mechanisms to enhance CNNs' feature representation capability, particularly by fusing richer contextual information or extracting global features to compensate for the limitations of traditional CNNs in capturing complex RS scene information.

B. LU Classification Based on CNN–Transformer Hybrid Networks

Beyond methods that enhance CNN performance, integrating CNNs with the Transformer architecture has emerged as another significant direction. This hybrid model aims to combine the strengths of CNNs in extracting local features with the Transformer's ability to capture global dependencies, thereby achieving more powerful scene understanding and improved classification accuracy. Yang et al. [48] proposed an efficient local–global lifting Transformer (LGLFormer) framework. The LGLFormer achieves this by effectively integrating CNNs, Transformers, and wavelet transforms to learn robust local–global features. Notably, it proposes unique LG adaptive feature extraction and LG lifting schemes to capture both local and global dependencies, significantly boosting the feature representation capabilities of CNNs for superior LU classification performance. Wang et al. [49] proposed an RS lightweight network with a CNN–Transformer hybrid framework, RingMo-Lite. Building upon the CNN–Transformer hybrid framework, this model leverages the frequency-domain properties of RS images. It treats the Transformer module as a low-pass filter and the CNN module as a stacked high-pass filter to effectively extract local and global features, achieving excellent performance across various RS image interpretation tasks. To effectively leverage the local–global features extracted by CNNs and Transformers, a new interactive dual-branch network was proposed in [50]. In the core two-way feature interaction module (TFIM), the CSwin→ResNet submodule is used to fuse global features into local features, aiming to blend channel information and enhance nonlinearity. Conversely, the ResNet→CSwin submodule extracts multiscale features through multiscale pooling multihead self-attention (MSPMHSA) and fuses local features into global features, thereby effectively capturing the multiscale local and global information present in RS scenes. In existing research, leveraging multiscale and multilevel features within CNN–Transformer architectures to capture both local details and global context has become a dominant paradigm.

Similarly, RMFFNet proposed by Wang et al. [51], PSCLI-TF by Li et al. [52], and the HCFPN network [53] all utilize CNN–Transformer hybrid architectures with multiscale and multilevel features to improve LU classification performance. Yuan et al. [54] proposed GLA-ViT, which enhances its ability to capture details and contextual information in RS images by integrating local and global self-attention modules. Addressing ViT's insufficient local feature capture in RS image scene classification, Wang et al. [55] introduced P^2FEViT, a hybrid

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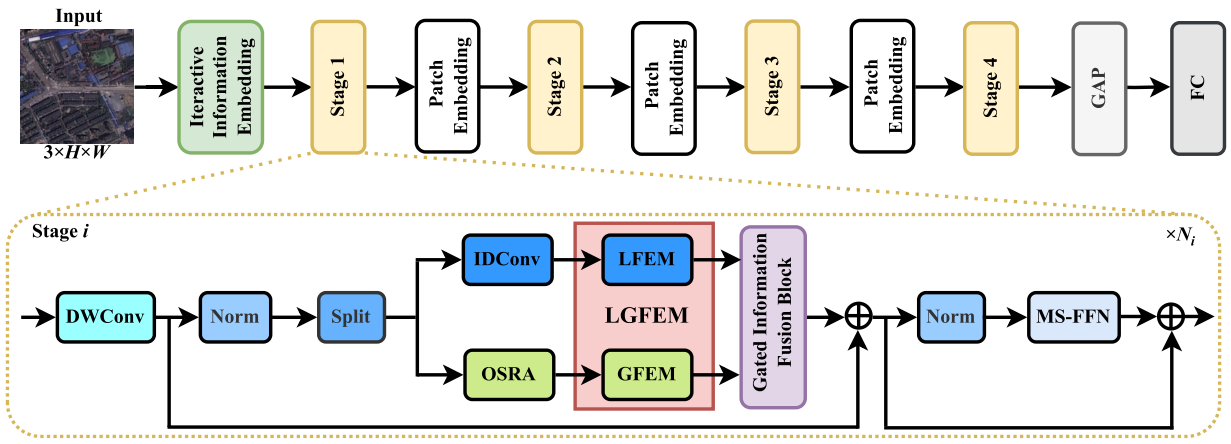


Fig. 1. Overall network structure.

architecture that integrates plug-and-play CNN features. This model aims to simultaneously capture and fuse global context with local multimodal information.

III. Proposed Method

In this section, the proposed model GLIFFM is presented in detail. The overall framework of the GLIFFM is introduced first, followed by the important components of the network.

A. Overview

As shown in Fig. 1, the proposed GLIFFM consists of three parts: interactive information embedding, global and local feature extraction, and gated feature fusion block. Our designed interactive information embedding serves as the initial patch embedding layer; its role is to incorporate LC primitives and facilitate the interactive fusion of LC and optical information, which can obtain complementary semantic features in addition to visual features. More importantly, it can emphasize the correlation and complementary property between optical and LC primitives for LU classification. Subsequently, the extracted features are fed to the employed OSRA and IDConv modules to acquire global and local features. Following this, the initial global and local features are enhanced via the local-feature enhancement block (LFEB) and global-feature enhancement block (GFEB). Finally, in order to facilitate the interaction and fusion of the enhanced global and local features, a gated information fusion block (GIFB) is proposed.

B. Interactive Information Embedding

The primary distinction between interactive information embedding and traditional patch embedding lies in its incorporation of LC primitives and the design of an MFE module. This architecture is specifically designed to overcome the limitation of traditional embedding methods, which often overlook the synergistic potential between LC semantic priors and raw optical textures. The MFE module aims to fuse LC and optical information to derive complementary features, thereby providing the subsequent global–local feature enhancement and fusion modules with a more semantically rich and interactive feature representation between optical and

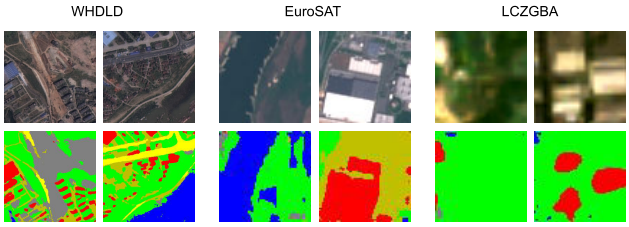


Fig. 2. Visual examples of the LC maps generated by U-Net++ across different datasets.

LC domains. By leveraging nested and dense skip pathways, U-Net++ effectively bridges the semantic gap across feature maps, facilitating accurate boundary delineation and multiscale feature extraction. Specifically, in the interactive information embedding layer, first, a pretrained U-Net++ network was fine-tuned to generate LC maps with optical images as inputs, serving as semantic-level information. The resulting LC maps generated across various datasets are visualized in Fig. 2. Both original optical images and generated LC maps are then fed into a dual-branch independent patch projection, converting them into Transformer-compatible, dimensionally consistent, serialized tokens that retain local features. These tokens represent the optical domain and LC features, respectively, and their interactive fusion is performed by the proposed MFE.

The overall architecture of MFE is shown in Fig. 3. It takes optical and LC features as input, which are then divided into four subfeatures, i.e., X_1–X_4. Specifically, X_1 and X_2 are derived from the optical imagery, whereas X_3 and X_4 represent the corresponding LC features, each with channels of $C/2$. Each subfeature is refined through a sequential convolutional block. First, a 3×3 convolution compresses the subfeature to $C/4$ channels, allowing it to learn more compact and potentially abstract feature representations. A Sigmoid function then normalizes it, and a second 3×3 convolution restores the subfeature to $C/2$ channels, yielding purified information. Finally, ReLu is introduced to help the module learn more complex features. The resulting refined subfeatures (i.e., X_1'–X_4') contain more distilled and profound features for the optical domain and LC information fusion. This process

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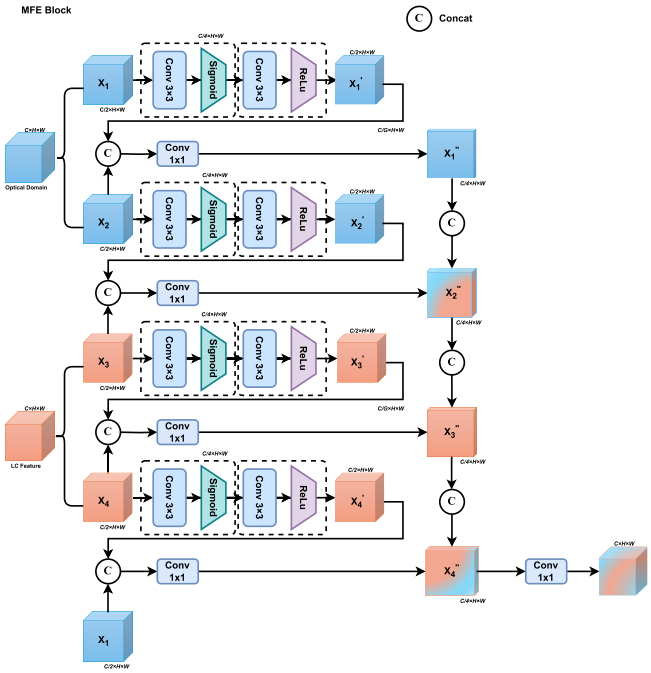


Fig. 3. MFE module.

is defined as follows:

$$X_i' = ReLu(Conv3(Sigmoid(Conv3(X_i)))). \tag{1}$$

In order to combine original but less abstract information with transformed and refined information, each refined and original subfeature pair, i.e., $\{X_i, X_{i+1}\}$ ($i = 1, 2$, and 3) and $\{X_4, X_1\}$, is separately concatenated and fed into a 1×1 convolution, represented as follows, with corresponding output denoted as X_i'':

$$X_i'' = Conv1(X_i' \oplus X_{i+1}). \tag{2}$$

The 1×1 convolution reduces the channel dimension of the concatenated features (which have C channels) to $C/4$. In this way, deeper-level features are retained, and excessive smoothing or loss of original details is avoided at the same time. More importantly, interrepresentation interaction in the optical domain and LC features is captured by X_1'' and X_3'', respectively. Intrarepresentation interaction across optical and LC features is learned via X_2'' and X_4''. Finally, these features X_i'' are concatenated and fed into a 3×3 convolution for further representation learning.

C. Local and Global Feature Enhancement

Inspired by the dual dynamic token mixer [56], input-dependent depthwise convolution (IDConv) and overlapping spatial reduction attention (OSAR) are employed for local and global feature extraction, respectively. In [56], upon concatenated local and global features, a squeezed token enhancer (STE) module, comprising a 3×3 depthwise convolution, two 1×1 convolutions, and a residual connection, is used for cross-channel communications. However, the STE module's focus on solely spatial local feature extraction is often suboptimal, compromising the model's ability to perceive the distribution of multiple LC primitives in LU classification tasks.

To address this limitation, we propose the local and global feature enhancement module (LGFEM) to enrich global context clues and local spatial contextual correlations for both interrepresentation and intrarepresentation interaction features. The design of LGFEM is specifically motivated by the necessity to simultaneously refine fine-grained optical textures and macrolevel LC distributions, ensuring that the model maintains a comprehensive perception of both raw image details and semantic spatial arrangements. LGFEM comprises two distinct branches: the local branch and the global branch. These branches receive features directly from the IDConv and OSRA modules, respectively. Integrated within each stage of the TransXnet backbone, the LGFEM operates across multiple scales by directly processing the feature maps generated at every hierarchical level.

Specifically, in the local feature enhancement branch, it consists of two pathways. The specific details of the local branch are illustrated in Fig. 4. It takes a local feature map $L \in \mathbb{R}^{H\times W\times C}$ as input.

In the first pathway, to obtain an spatial attention map that can be aware of representation of important areas in image scenes, we choose to concatenate max-pooling ($MaxPool$) and average-pooling ($AvgPool$) features along the channel dimension, and feed them into a 3×3 convolution followed by a Sigmoid function, according to specific enhanced feature fusion module (SEFFM) in [57]

$$L_1^{concat} = Concat(MaxPool(L), AvgPool(L)) \tag{3}$$

$$L_1^{sam} = Sigmoid(Conv3(L_1^{concat})) \tag{4}$$

where $L_1^{concat} \in \mathbb{R}^{H\times W\times 2}$ and Concat $(\cdot)$ denotes the concatenation operation.

In the second pathway, we first divide L along the channel dimension into two subfeature maps, termed $\{L_1, L_2\} \in \mathbb{R}^{H\times W\times(C/2)}$. Dual parallel convolutional groups take L_1 and L_2, respectively, to further feature enhancement.

In the first convolutional group, L_1 undergoes refinement and channel upsampling using a 3×3 convolution, followed by a nonlinear combination of batch normalization and ReLu to enhance its expressive power. A second 3×3 convolution then downsamples the channels back to the original count, with GeLu providing a richer nonlinear mapping. The second convolutional group is reconfigured by entirely replacing all 3×3 convolutions with 1×1 convolutional operations. This provides a lightweight local feature processing pathway that focuses on interchannel relationships, enabling efficient channel mixing and nonlinear transformations without expanding the receptive field. Enhanced L_1 and L_2 features are then concatenated as $L' \in \mathbb{R}^{H\times W\times C}$. By utilizing the second pathway to process the original local feature L, more comprehensive and richer local information from diverse perspectives can be obtained through parallel convolutional groups processing by emphasizing both spatial locality and channel mixing.

The two-patch merging is achieved by elementwise multiplication of the attention map from the first pathway with the features obtained from the second pathway. It should be noted that the spatial attention map is obtained based

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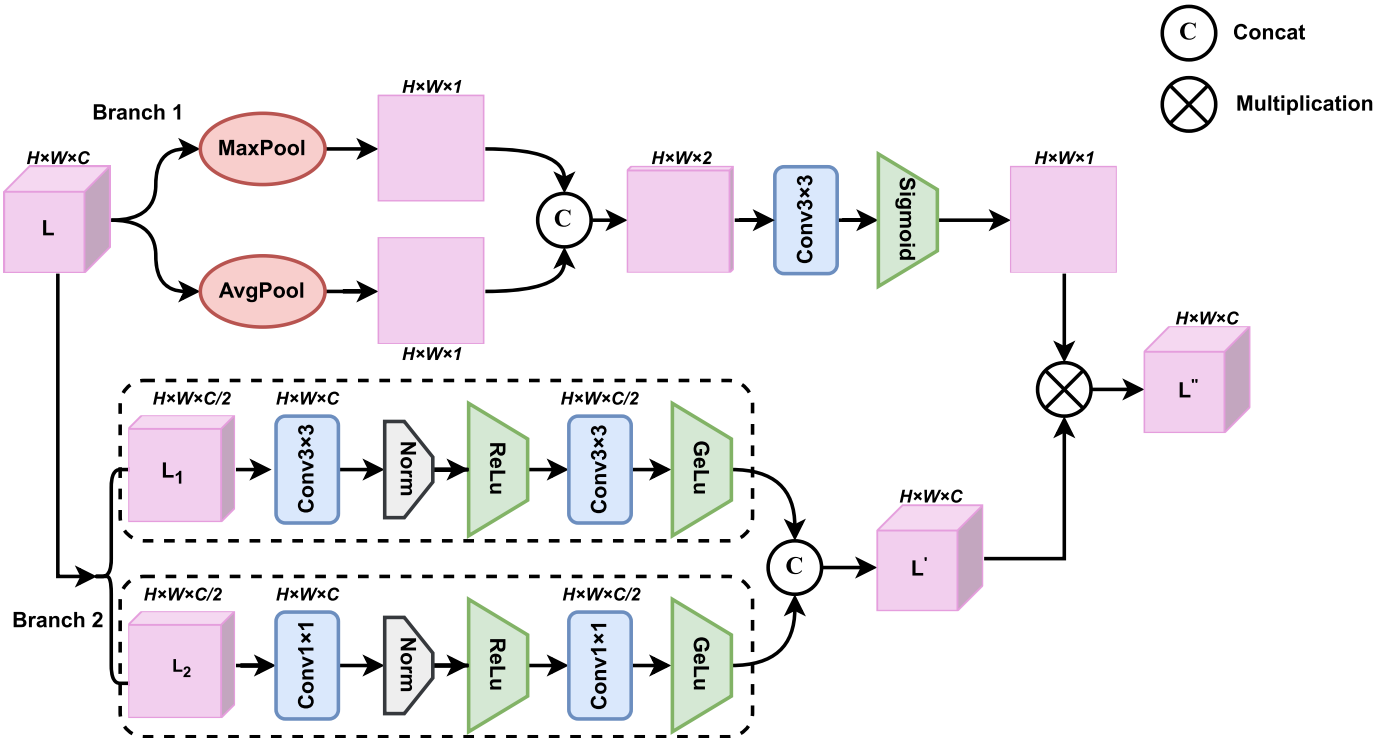


Fig. 4. Local feature enhancement branch.

on the original local feature L but not the refined feature L' due to the fact that L contains more spatially detailed feature information compared to L', allowing it to judge which regions are important from an original and global perspective. Multiplying this obtained attention map by the refined and enhanced local semantic features (i.e., L') can achieve optimal weighting to selectively enhance the most contributive and refined local semantic information. Overall, the role of the local branch is to weigh the high-level local features and optimize the local representation before fusion. The processed local feature L'' thus contains richer, more abstract, and highly discriminative high-level local semantic information.

For the global feature enhancement branch, it also consists of two pathways, as illustrated in Fig. 5. In the first pathway, a similar structure as SEFFM in [57] is adopted to obtain channel attentiveness global features. Specifically, for the global feature map $G \in \mathbb{R}^{H\times W\times C}$, each feature channel is fed to parallel max and average pooling. Subsequently, the two pooling results were sent into two parallel weights-unshared bottleneck convolution blocks (B Conv), separately, unlike a shared block in [57]. Separate convolutional groups allow customized processing for max and average pooling results, eliminating shared-weight constraints on their statistical differences. Each BConv consists of two 1×1 convolutions, which downsample the channels to $C/4$ and then expand them back to $C/2$, with ReLU activations between the two 1×1 convolutions.

This modified architecture affords enhanced flexibility and nonlinear expressive capacity, empowering the model to capture complex cross-interactions between the two information types, rather than merely implementing a simple addition followed by separate activations. Finally, we perform elementwise multiplication between these refined weights and the original global feature map G, yielding the channel-attention-weighted global feature representation G_1.

In the second pathway, we first perform global average pooling on G, then flatten the pooling result and send it into a global nonlinear semantic encoder to capture deeply abstracted global semantic representations. This encoder comprises two fully connected layers, each immediately followed by a GeLu activation function, designed to perform high-level nonlinear transformations on the aggregated contextual information. Subsequently, the transformed features are unflattened to match the spatial dimension of the original input, yielding the output of this pathway, i.e., the abstracted semantic representation G_2.

Comparing the outputs of the dual pathways in the global feature enhancement branch (i.e., G_1 and G_2), several fundamental distinctions are observed. G_1 retains the spatial structure of original features while performing channelwise importance recalibration, focusing on dynamic modulation of raw global features. In contrast, G_2 discards spatial details to concentrate on distilling higher-order abstract semantics from globally averaged information. It encodes the contextual information of the entire scene through deep nonlinear transformations, thereby providing profound semantic comprehension of the global scene.

The global branch thus constructs two parallel information processing flows: one maintains spatial fidelity with channel-adaptive refinement (G_1), and another extracts translation-invariant semantic concepts (G_2). These complementary representations are ultimately integrated through an

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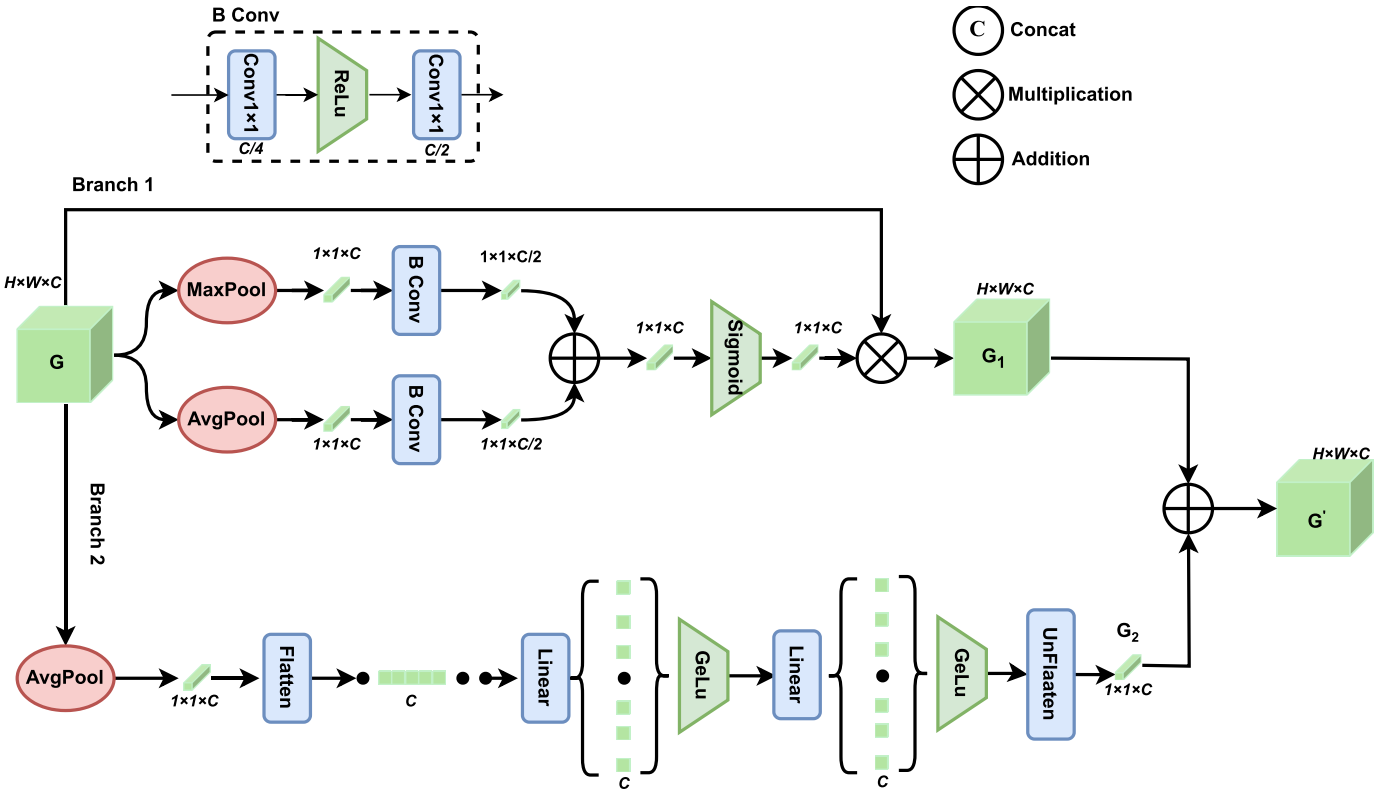


Fig. 5. Global feature enhancement branch.

addition operation, synergistically combining G_1's feature-recalibration properties with G_2's contextual abstraction to form a comprehensive and robust global feature representation (i.e., G'). This fused output provides the foundation for subsequent integration with refined local features in the local feature enhancement branch.

D. Gated Information Fusion Block

GIFB is developed with the aim of fusing refined local and global features (i.e., L'' and G') generated by the LGFEM, as depicted in Fig. 6. The inclusion of this block is primarily intended to handle the complex nonlinear dependencies between optical textures and semantic priors, which simple concatenation or addition cannot adequately address.

Initially, L'' and G' are concatenated along the channel dimension to produce a fused feature F with $2C$ channels. Subsequently, F is partitioned along the channel axis into g feature groups, each with the size of $H \times W \times 2C/g$. The grouping strategy permits learning of diverse fusion policies on refined feature subspaces. Each grouped feature F_i undergoes a 3×3 convolution layer with an output channel of 1 and a Sigmoid layer for normalization to create a single-channel spatial weight map. The normalized attention maps are then elementwise multiplied with their corresponding grouped features, implementing a gating mechanism that produces weighted feature representations. The gated grouped features are reassembled through concatenation. Subsequently, the concatenated features are refined via a 3×3 convolution to aggregate information across channels and local contexts. Finally, an additive residual connection is added by combing output of the 1×1 convolution with its input, enhancing information preservation.

The group-gating fusion mechanism within this block facilitates the interaction and fusion of local and global subfeatures in different ways. Due to the intricate feature relationships where local and global features do not merely coexist, they can be actively collaborated via mutual interaction and adaptation by gating mechanisms, collectively governing the ultimate fusion output. Since gated weights are derived from each grouped feature F_i, which encompasses both local and global information, their contributions are codetermined during feature integration.

IV. Experiment and Results

The proposed network for LU classification will be examined in this section. First, we introduce the datasets and evaluation metrics used in our experiments. Next, the method's performance was compared with state-of-the-art techniques, and model complexity was analyzed. At last, ablation studies were performed on three datasets to assess the effectiveness of the proposed method.

A. Dataset

To evaluate the performance of our model, we conducted experiments on three open datasets of satellite RS imagery, including WHDLD [39], [40], EuroSAT [58], [59], and LCZ–Guangdong–Hong Kong–Macao Greater Bay Area (LCZGBA) [60]. Table I provides the number of images

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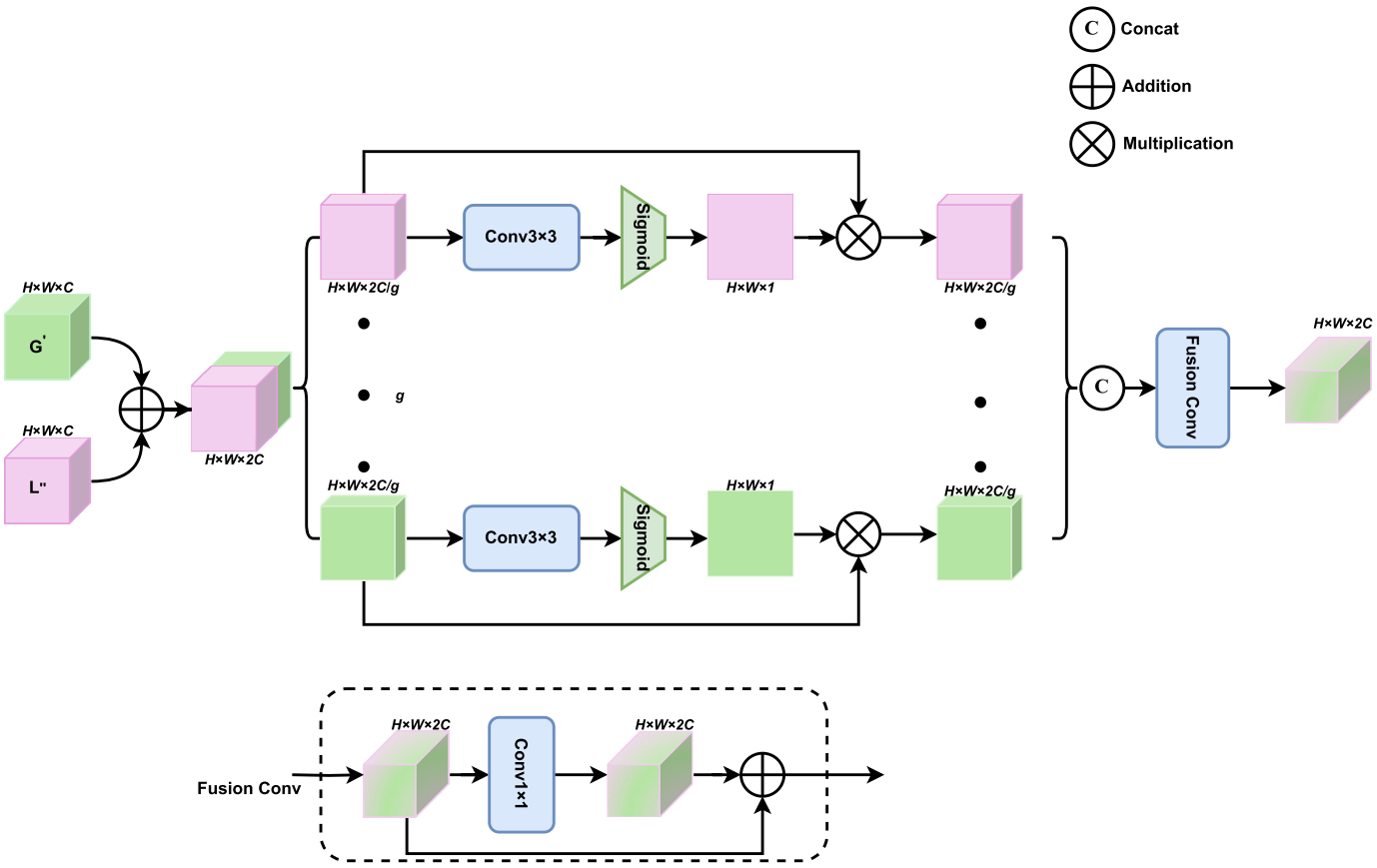


Fig. 6. GIFB.

present in the three datasets for each LU category. Specifically, all datasets are partitioned into training and testing sets based on predefined ratios, where the former is utilized to optimize the LU classification model parameters, while the latter serves to independently evaluate final performance and generalization capability.

WHDLD is a dense annotation dataset for semantic segmentation, with a total of six LC categories at the pixel level, namely buildings, roads, sidewalks, vegetation, bare soil, and water. WHDLD with pixel-level LC labels is used to train the U-Net++ network, and the trained model serves to predict LC primitives for the three datasets. LU types were identified and labeled manually through visual inspection by combining geographic locations from Google Maps and the RS image. A total of 4940 images in WHDLD, each with a size of 256 × 256 pixels, were manually classified into ten LU categories for LU dataset construction. Some example images with the corresponding pixel-level LC labels and patch-level LU labels of the WHDLD dataset are shown in Fig. 7(a).

EuroSAT contains 27 000 images taken from Sentinel-2 satellite across Europe. It consists of ten major land-use types with 2000 to 3000 image patches per LU category, as summarized in Table I. Each image has a size of 64×64 pixels and a spatial resolution of 10 m. The EuroSAT optical dataset is used in this study. Example images of the EuroSAT dataset are shown in Fig. 7(b).

The LCZGBA dataset comprises eight built types and six land-cover types, detailed in Table I. The dataset images were acquired from the Landsat 8, which has a spatial resolution of 30 m. The available LCZ label map is at a 100-m resolution. Following [60], the images were bilinearly resampled to a 10-m resolution. To generate a patch-level dataset and deal with the spatial-resolution mismatch between images and label map, the images were cropped into 10 000 patches with a size of 60×60 pixels, and the LCZ type for each path was identified as a majority type exceeding 70% in a sliding window. Examples of the LCZGBA dataset are shown in Fig. 7(c).

B. Implementation Details and Evaluation Metrics

All experiments were implemented in PyTorch on an NVIDIA GeForce RTX A6000 GPU with 48 GB memory. In the GIFB module, the number of groups g was empirically set to 4. The Adam optimizer was employed with a weight decay of 0.0001, utilizing cross-entropy loss as the loss function. The initial learning rate was set to 0.0001, utilizing a cosine-decay learning-rate scheduler with linear warm-up. Considering the varying dataset sizes, the batch sizes for the WHDLD, Eurosat, and LCZGBA datasets were set to 64, 128, and 128, respectively. In addition, basic data augmentation techniques, including random rotations and horizontal and vertical flipping, were applied.

For model training, we randomly select 80% and 50% of the images to train both the U-Net++ and LU classification

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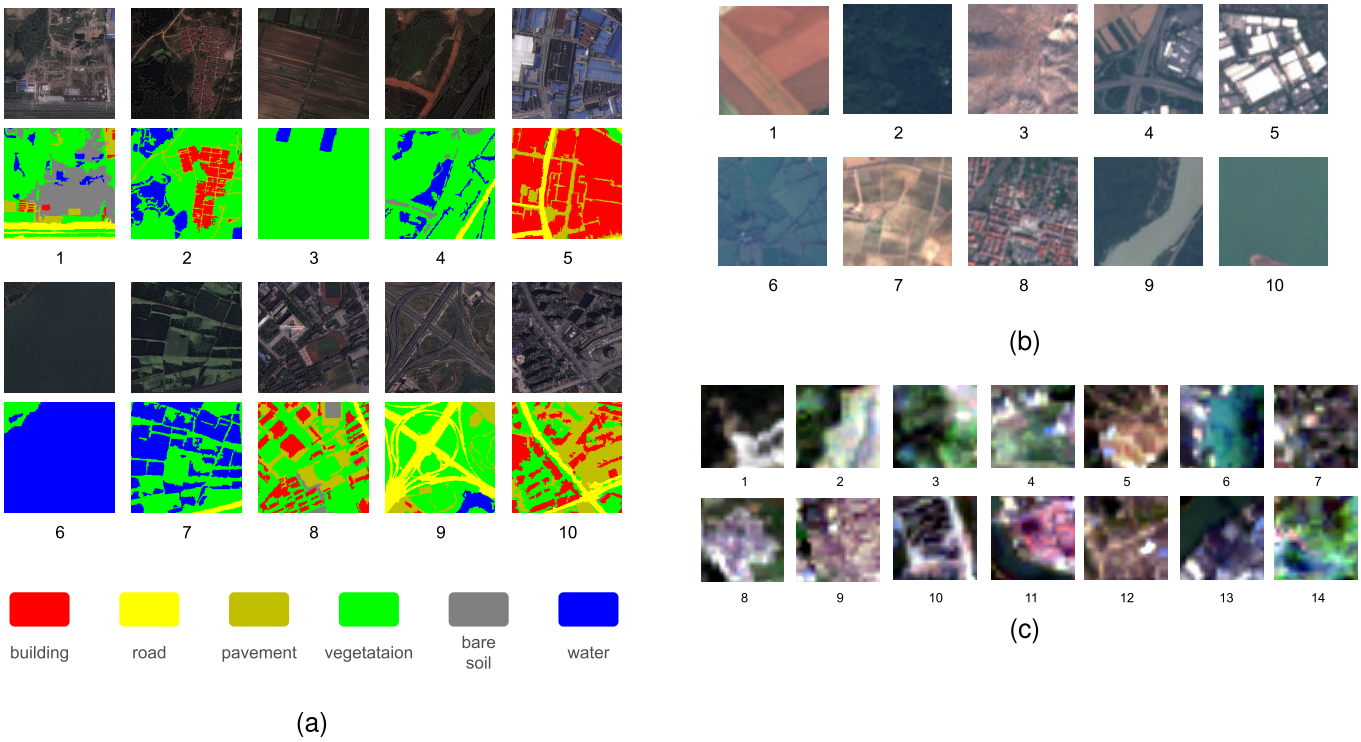


Fig. 7. (a) WHDLD, (b) EuroSAT, and (c) LCZGBA. The LU category ID can be referred to Table I.

TABLE I
DESCRIPTION OF SELECTED DATASETS FOR THE CLASSIFICATION OF LU FROM RS IMAGES

WHDLD		EuroSAT		LCZGBA	
Class name	# Images	Class name	# Images	Class name	# Images
1. Construction sites	424	1. Annual Crop	3000	1. Large low-rise	869
2. Country house	461	2. Forest	3000	2. Scattered trees	216
3. Cropland	160	3. Herbaceous Vegetation	3000	3. Bare soil or sand	627
4. Greenbelt	560	4. Highway	3000	4. Low plants	947
5. Industrial land	637	5. Industrial	2500	5. Bare rock or paved	366
6. Natural waters	941	6. Pasture	2000	6. Water	1716
7. Pond	199	7. Permanent Crop	2500	7. Compact mid-rise	272
8. School	161	8. Residential	3000	8. Compact low-rise	132
9. Transpor tation facilities	327	9. River	2500	9. Compact high-rise	38
10. Urban dwellings	1070	10. Sea Lake	3000	10. Open high-rise	463
Total	4940	Total	27500	11. Open low-rise	413
				12. Open mid-rise	389
				13. Heavy industry	90
				14. Dense trees	4028
				Total	10566

models in the WHDLD. While for EuroSAT with a relatively larger number of samples (i.e., 27 000 images), two training ratios of 50% and 20% are used to obtain two subsets. For LCZGBA, the training ratio is set to 50%. For evaluation, we used three widely used evaluation metrics, overall accuracy (OA), Kappa coefficient, and confusion matrix. OA measures the proportion of samples that the model correctly classifies. Considering the unbalanced number of samples of different LU categories, especially in the WHDLD and LCZGBA, Kappa, which measures the classification consistency and considers the consistency caused by chance were used. In addition, the confusion matrix is introduced to visualize class-specific performance, including classwise precision and interclass classification error.

C. Comparison With State-of-the-Art Methods

To thoroughly evaluate the effectiveness of our proposed method, a comprehensive comparative analysis was conducted against a selection of recent state-of-the-art models. The chosen benchmarks can be categorized into two groups: general-purpose models, namely EdgeNeXt (base), PVT_V2 (B0), LeMeViT (Tiny), and StarNet (S2); and RS community-specific models, including RANet, EMTCAL (resnet34), MF^{2}CNet, SAGN (resnet34), and MSCN (resnet50). To ensure a fair and rigorous comparison, all models were implemented based on their respective authors' released codebases, with free parameters strictly following the original literature. Crucially, all experiments were performed under the identical settings outlined in Section IV-B, and no pretrained weights were utilized for any of the methods.

1) Results of the WHDLD Dataset: The performance metrics for the WHDLD dataset under different training ratios (8:2 and 5:5) are summarized in Table II. It is evident that GLIFFM consistently outperforms competing methods across both split scenarios. Under the 8:2 split, GLIFFM achieves an OA improvement of 0.61% over EMTCAL and 1.31% over

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Fig. 8. Confusion matrices for the GLIFFM model on the EuroSAT dataset. The corresponding semantic names for the numbered classes are listed in Table I. (a) 80% training ratio. (b) 50% training ratio.

TABLE II
COMPARISON OF OAS (%) AND KAPPA (%) ON THE WHDLD DATASET

Methods	Training rates			
	80%		50%	
	OA	Kappa	OA	Kappa
RANet [65]	70.26	65.09	68.50	63.09
EMTCAL [66]	76.51	72.67	74.36	70.10
MF^{2}CNet [67]	74.60	70.36	73.35	68.95
EdgeNeXt [61]	69.46	64.08	67.89	62.26
PVT_V2 [62]	72.58	68.12	70.20	65.13
SAGN [68]	75.81	71.90	74.04	69.84
LeMeViT [63]	74.50	70.37	72.50	67.88
StarNet [64]	71.77	67.18	69.96	64.80
MSCN [69]	72.88	68.42	72.83	68.46
Ours	**77.12**	**73.34**	**74.48**	**70.31**

SAGN. Furthermore, even when the training ratio is reduced to 50%, GLIFFM also demonstrates leading performance with an OA of 74.48% and a Kappa coefficient of 70.31%. Compared to EMTCAL, our model achieves an improvement of 0.12% in OA and 0.21% in Kappa. This further validates that GLIFFM maintains a significant competitive edge even with a substantially reduced amount of training data.

The exceptional performance of our GLIFFM model on the WHDLD dataset can be attributed to two main factors. First, the incorporation of LC information provides the model with more comprehensive spatial–geographical and semantic data. This is particularly beneficial for complex scenes like the Transportation facilities category. In these cases, the presence of varied background elements, such as residential areas or cropland, can lead to misclassification by other models. Our GLIFFM addresses this challenge effectively, achieving the highest accuracy for the Transportation facilities category at 59% (under the 8:2 split). Second, the serial application of the LGFEM and GIFB modules plays a crucial role. The LGFEM module first enhances the features enriched with LC information before feeding them into the GIFB. The GIFB then filters and weights the most relevant LU and LC information, creating a more robust and informative fused feature representation. A prime example is the pond category, where our model achieved the highest accuracy of 70% (65% under the 5:5 split) among all models. Ponds typically consist of vegetation and water bodies with relatively regular arrangements. However, general-purpose models often struggle to distinguish ponds from Greenbelt and Cropland, as these categories can also contain water. Our method successfully handles this distinction. To further indicate the classification performance of our proposed network, the confusion matrix of GLIFFM is shown in Fig. 8. It is obvious that the misclassification primarily appears in the prediction of urban dwellings, especially for the misclassification of schools into urban dwellings (i.e., 0.45 and 0.28 under the 5:5 split), due to the similarity in RS scenes and class imbalance problem.

2) Results of the EuroSAT Dataset: To compare the proposed method with the comparative methods on the EuroSAT dataset deeply, two training datasets with training ratios of 20% and 50% are established. Similarly, the OAs and Kappa coefficient of the two EuroSAT training datasets are shown in Table III. The accuracy of GLIFFM obviously exceeds all the listed methods. In addition, most models misclassify the pasture category into annual crop and herbaceous vegetation when the training data is extremely limited (20% training ratio), due to the interclass similarity between them. However, GLIFFM stands out, achieving an accuracy of 91% for the pasture class. Besides GLIFFM, SAGN performed best with an accuracy of 88% for the pasture class. Fig. 9 shows the confusion matrices of the GLIFFM on the EuroSAT dataset. This result validates that GLIFFM can effectively improve classification performance for complex categories with limited datasets, due to the inclusion of LC primitives and its robust processing of interactive information.

3) Results on the LCZGBA Dataset: Even though the classification scheme in the LCZGBA dataset is the combination of built-up types and LC semantic classes, which is a little

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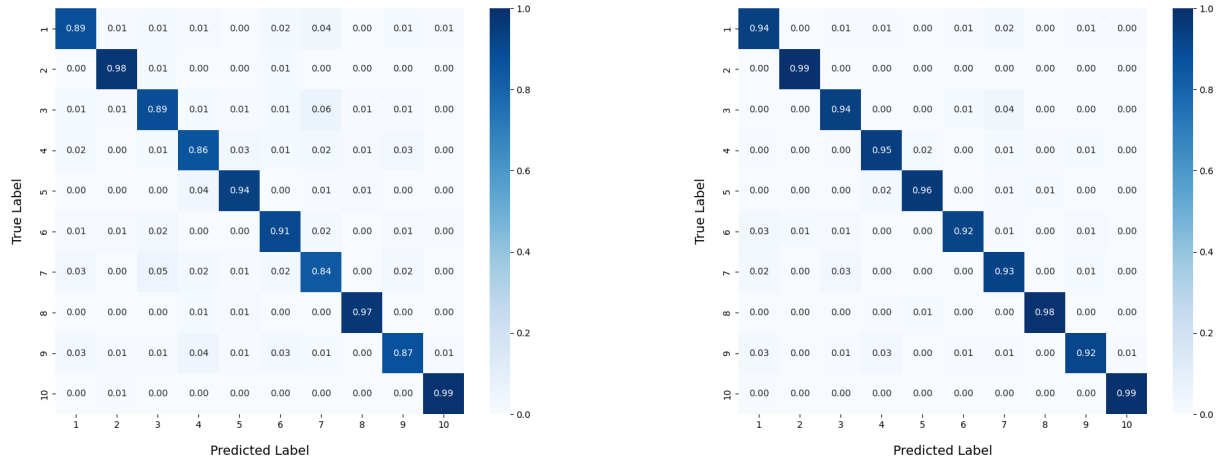


Fig. 9. Confusion matrices for the GLIFFM model on the EuroSAT dataset. The corresponding semantic names for the numbered classes are listed in Table I. (a) 20% training ratio. (b) 50% training ratio.

TABLE III
COMPARISON OF OAS (%) AND KAPPA (%) ON THE EUROSAT DATASET

Methods	Training rates			
	20%		50%	
	OA	Kappa	OA	Kappa
RANet [65]	81.40	79.30	87.69	86.29
EMTCAL [66]	88.26	86.93	94.07	93.40
MF^{2}CNet [67]	87.34	85.91	91.92	91.00
EdgeNeXt [61]	81.94	79.89	88.88	87.62
PVT_V2 [62]	82.07	80.04	88.32	87.00
SAGN [68]	88.74	87.47	93.93	93.25
LeMeViT [63]	87.29	85.86	92.81	92.00
StarNet [64]	85.71	84.09	90.29	89.19
MSCN [69]	87.56	86.15	93.01	92.23
Ours	**91.61**	**90.66**	**95.43**	**94.91**

TABLE IV
COMPARISON OF OAS (%) AND KAPPA (%) ON THE LCZGBA DATASET

Methods	OA	Kappa
RANet [65]	71.02	63.52
EMTCAL [66]	76.13	70.21
MF^{2}CNet [67]	74.39	67.92
EdgeNeXt [61]	70.53	62.94
PVT_V2 [62]	75.58	69.35
SAGN [68]	76.24	70.34
LeMeViT [63]	75.71	69.64
StarNet [64]	73.10	66.28
MSCN [69]	76.56	70.67
Ours	**78.66**	**73.32**

different from the previous two LU datasets, the experimental results can testify to the applicability of the method in the LCZ classification task. The result of different methods is summarized in Table IV. From this table, we can find that the performance of GLIFFM is the best with the OA and Kappa of 78.66% and 73.32%, respectively. Given the high degree of class imbalance and low resolution of this dataset, the classification task is particularly challenging. Despite this, our method still achieved promising results. Compared with the other

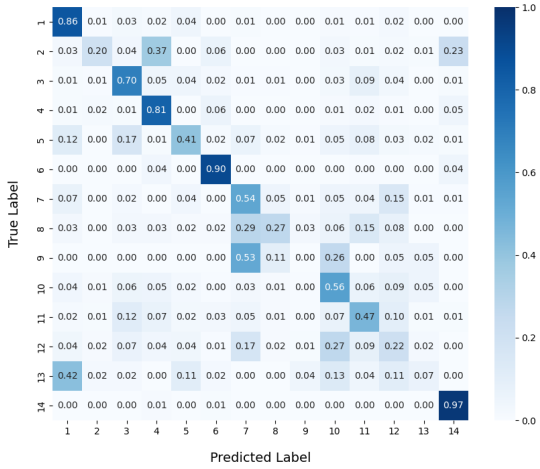


Fig. 10. Confusion matrix for the GLIFFM model on the LCZGBA dataset is presented. The corresponding semantic names for the numbered classes are listed in Table I.

models, the performance enhancements in Kappa achieved by GLIFFM are 9.80% over RANet, 3.11% over EMTCAL, 5.4% over MF^{2}CNet, 10.38% over EdgeNeXt, 3.97% over PVT_V2, 2.98% over SAGN, 3.68% over LeMeViT, 7.07% over StarNet, and 2.65% over MSCN.

Besides, the confusion matrix of our GLIFFM is exhibited in Fig. 10. Due to the visual similarity between dense trees, low plants, and scattered trees, and the presence of vegetation-like backgrounds in some open mid-rise samples in the low-resolution LCZGBA dataset, it is particularly difficult to distinguish among these categories. Consequently, all the models perform poorly on these categories with significant misclassification among the three tree-related classes. Addressing this limitation and further improving GLIFFM's performance are our future works. However, it is encouraging that our method outperforms all comparison models in the open low-rise buildings category, which further validates the performance of GLIFFM.

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TABLE V
COMPARISON OF PARAMETERS AND FLOPS OF DIFFERENT METHODS ON THE WHDLD, EUROSAT, AND LCZGBA DATASETS

Methods	Dataset					
	WHDLD		EuroSAT		LCZGBA	
	FLOPs (G)	Parameters (M)	FLOPs (G)	Parameters (M)	FLOPs (G)	Parameters (M)
RANet [65]	9.09	26.28	0.57	26.28	0.57	26.28
EMTCAL [66]	5.53	27.29	0.35	27.29	0.35	27.30
MF2CNET [67]	3.17	26.02	0.20	26.02	0.20	26.84
EdgeNeXt [61]	3.81	17.91	0.24	17.91	0.24	17.91
PVT_v2 [62]	0.70	3.41	0.04	3.41	0.04	3.41
SAGN [68]	7.98	22.77	0.50	22.77	0.50	22.78
LeMeViT [63]	2.30	8.32	0.25	8.32	0.25	8.32
StarNet [64]	0.99	5.50	0.06	5.50	0.06	5.50
MSCN [69]	8.39	36.25	0.52	36.25	0.52	36.26
U-Net++ (LC Stream)	18.50	26.08	1.16	26.08	1.16	26.08
Ours	3.16	16.10	0.20	16.10	0.20	16.10

D. Model Complexity Analysis

Beyond the evaluation of performance metrics, like OA and Kappa, we also assess the computational efficiency of the models by analyzing their parameter counts and FLOPs. Such indicators are of significant importance for deployment in resource-constrained scenarios.

As presented in Table V, our approach stands out by offering a pragmatic and robust solution for LU classification. While lightweight models like PVT_v2 and StarNet demonstrate minimal computational costs, their suboptimal performance on these three specific datasets limits their practical utility. In contrast, leading RS methods such as EMTCAL and SAGN, despite their strong results, incur substantially higher computational burdens. Our method bridges this gap by providing strong performance with significantly fewer parameters and FLOPs, making it a more practical choice for real-world deployment.

As our system follows a two-stage pipeline, the total inference cost includes both LC map generation and LU classification. To provide a transparent analysis, we report the overhead of the U-Net++ module (with a ResNet-34 backbone) in Table V. For a standard input size of 256×256, U-Net++ introduces 26.08 M parameters and 18.50 G FLOPs. When the input is scaled down to 64×64 for specific processing requirements, the computational demand of U-Net++ decreases to 1.16 G FLOPs, while the parameter count remains constant.

E. Ablation Experiments

In this section, to analyze the contributions of different components, including MFE, LGFEM, and GIFB, within our GLIFFM, we construct the ablation experiments on three datasets.

To this end, we construct the following nets.

1) Net_0: Baseline.
2) Net_1: Baseline + MFE.
3) Net_2: Baseline + MFE + LGFEM.
4) Net_3: Baseline + MFE + LGFEM + GIFB.

Here, Net_0 is TransXnet-Tiny [56], Net_1 is built upon Net_0 with the addition of MFE. Net_2 incorporates LGFEM to enhance local and global features before the feature fusion of Net_1. Net_3 is our GLIFFM. For Net_0–Net_2, the fusion of local and global features is achieved using TransXnet's STE module, while Net_3 substitutes STE with our proposed GIFB. The results counted on three datasets are shown in Fig. 11, and we can get the following observations. First, all of our modules have a positive impact on the classification performance. Second, the performance of Net_0 is the weakest among the four nets. When introducing the proposed MFE, Net_1 consistently outperforms Net_0 across all datasets, and its performance gain over the previous network is the highest among all pairs, which illustrates that introducing LC using the MFE module has a significant positive effect on LU classification. Third, the introduction of LGFEM in Net_2 resulted in superior performance over Net_1 across all datasets, validating its effectiveness. These results also indicate that our dual-branch local–global feature processing method significantly enhances feature representation and boosts the model's discriminative power. At last, after replacing STE with GIFB in Net_2, Net_3 achieves the best classification results. This demonstrates that GIFB is better suited for local–global feature fusion tasks when integrating LC features. In addition, we will provide a detailed analysis of each component within the GLIFFM.

1) Effectiveness of MFE: As mentioned in Section III-B, the MFE serves to introduce LC primitives and fuse them with optical features, thereby adding spatial semantic information to LU scenes, which is especially beneficial for complex scenes or categories that are difficult to classify due to their spectral characteristics. The benefits of MFE are particularly evident on the data-rich EuroSAT dataset, where its relatively balanced classes allow the injected LC information to significantly improve the classification of categories with high intraclass variance and low interclass variance. On the other hand, the high classification difficulty of the LCZGBA dataset makes it challenging for visual information alone to discern the LCZ features from low-resolution imagery. In this case, the LC information extracted by MFE effectively compensates for the lack of visual details due to the low resolution.

As depicted in Fig. 11, Net_1 consistently outperforms Net_0 on all datasets. An analysis of the computational efficiency

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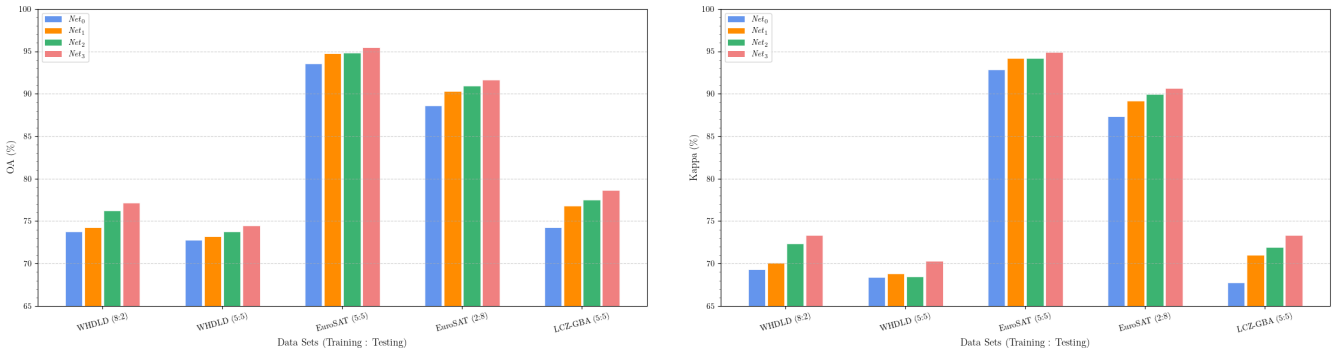


Fig. 11. OA and Kappa values of Net_0–Net_3 on the WHDLD, EuroSAT, and LCZGBA datasets. (a) OA and (b) Kappa.

TABLE VI
COMPARISON OF PARAMETERS AND FLOPS OF Net_0–Net_3 ON THE WHDLD, EUROSAT, AND LCZGBA DATASETS

Methods	Dataset					
	WHDLD		EuroSAT		LCZGBA	
	FLOPs (G)	Parameters (M)	FLOPs (G)	Parameters (M)	FLOPs (G)	Parameters (M)
Net_0	2.31	11.88	0.15	11.88	0.15	11.88
Net_1	2.50	11.92	0.16	11.92	0.16	11.92
Net_2	3.03	15.31	0.19	15.31	0.19	15.31
Net_3	3.16	16.10	0.20	16.10	0.20	16.10

in Table VI indicates that despite the notable performance improvement, Net_1 only adds 0.04 M parameters relative to Net_0. More importantly, on the smaller EuroSAT and LCZGBA datasets, this performance boost is achieved with an almost imperceptible FLOPs increase of just 0.01 G. This demonstrates that our MFE module effectively balances performance enhancement with computational efficiency, providing significant gains with minimal computational cost.

2) Effectiveness of LGFEM: LGFEM is a dual-branch feature enhancement module designed to strengthen global–local features enriched with LC information. Its effectiveness is particularly evident on datasets like WHDLD, which are characterized by sample imbalance, a small number of samples, and complex background classes, such as school and transportation facilities. For these categories, classification relies on more than just LC and optical information, and it requires stronger spatial features to distinguish the primary LU information from redundant and complex backgrounds. LGFEM's feature enhancement capability addresses this by further refining the deep features infused with LC information, resulting in more discriminative features that can effectively separate LU categories from their complex backgrounds.

Although the integration of LGFEM increases the parameter count by 3.39 M relative to Net_1, the benefits are considerable. The module provides a substantial performance boost, especially for datasets characterized by limited samples and complex backgrounds. For our future work, we intend to investigate more lightweight enhancement strategies to reduce model complexity while maintaining performance.

3) Effectiveness of GIFB: As shown in Fig. 11, all datasets show a substantial improvement after integrating GIFB, which validates that our GIFB is more suitable for the LU classification task than STE. The GIFB, by using a group gating mechanism, effectively filters the local–global information output by LGFEM and then concatenates it. This approach not only preserves important LU information but also mitigates the interference of peripheral background information on the classification task. While the performance of GIFB is highly promising, we acknowledge that its effectiveness can be further enhanced. For instance, the grouped gating mechanism may sometimes lead to misclassifications of simple, easily distinguishable categories and introduce a high computational overhead. Our future work will therefore focus on exploring more advanced gating strategies to better handle these issues and address parameter redundancy. This will further improve GIFB's performance and efficiency, enabling the application on a wider range of challenging datasets.

V. DISCUSSION

A. Performance Comparison With ImageNet Pretraining

To further assess the robustness of GLIFFM under various initialization conditions, we conducted experiments using ImageNet-1K pretrained weights for all comparative models. This evaluation was carried out on the WHDLD (8:2) dataset.

The results are summarized in Table VII. It can be observed that the integration of pretrained weights leads to a performance improvement for all architectures. Notably, GLIFFM continues to outperform other state-of-the-art methods, achieving an OA of 79.33% and a Kappa coefficient of 75.98%. Specifically, compared to the best-performing baseline, SAGN, our method achieves a 0.05% higher Kappa. Furthermore,

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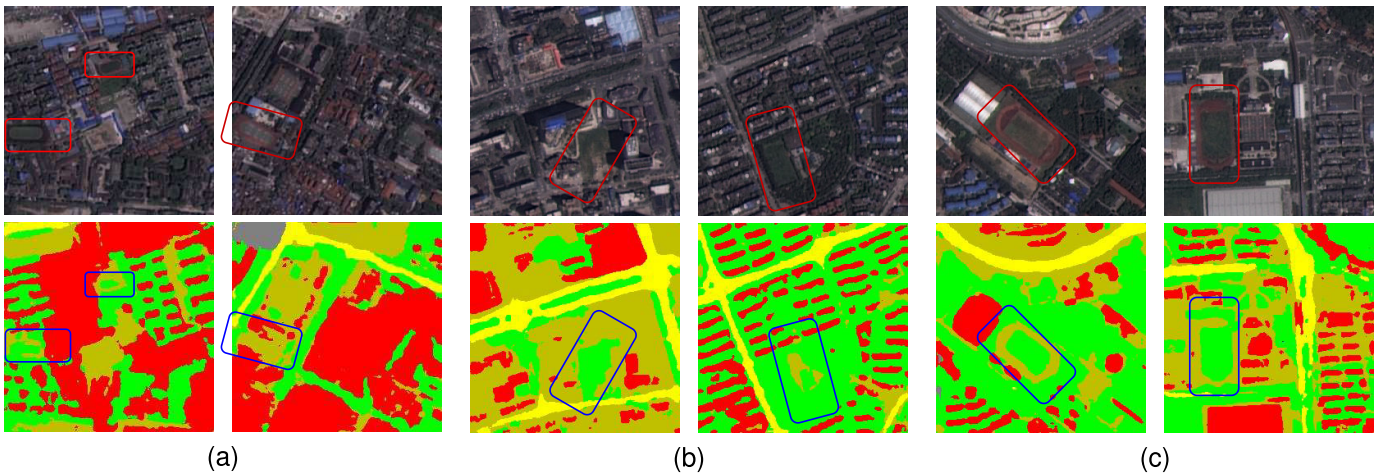


Fig. 12. Visual analysis of classification results on the WHDLD dataset. (a) Failure case of school. (b) Case of Urban dwellings. (c) Correct case of school. (Top) The original optical RS images. (Bottom) The corresponding LC maps predicted by U-Net++.

TABLE VII
EXPERIMENTAL RESULTS OAS (%) AND KAPPA (%) ON THE WHDLD (8:2) DATASET USING PRETRAINED WEIGHTS

Methods	OA	Kappa
RANet [65]	76.81	72.97
EMTCAL [66]	79.23	75.84
PVT_V2 [62]	78.93	75.46
SAGN [68]	**79.33**	75.93
StarNet [64]	79.23	75.77
MSCN [69]	79.03	75.62
Ours	**79.33**	**75.98**

when compared with EMTCAL, GLIFFM shows improvements of 0.1% in OA and 0.15% in Kappa. This indicates that the semantic guidance provided by the MFE module remains a significant driver of classification accuracy, regardless of whether or not pretrained features are utilized.

B. From LC Semantics or Interactive Features

Furthermore, to investigate whether the performance gain originates from true LC semantics or simple structural enhancement, we introduced a "Late Fusion" strategy for comparison. As shown in Table VIII, Late Fusion involves concatenating global LC category statistics with optical features at the classifier head. The experimental results across different datasets, summarized in Table VIII, demonstrate that our proposed interactive information embedding significantly outperforms Late Fusion. This quantitative evidence proves that the LC information provided by MFE does not merely act as a supplementary label; instead, its strength lies in providing spatially aligned semantic guidance at the embedding stage, which is essential for reconstructing discriminative features in complex LU scenes.

C. Impact of Grouping Granularity in GIFB Gating Mechanism

To further justify the selection of the group number g in the gating mechanism, we conducted additional ablation experiments with $g \in \{2, 4, 8\}$. The results, summarized in Table IX, show that $g = 4$ achieves the best balance between classification accuracy and model complexity. Specifically, while $g = 2$ increases the parameters and $g = 8$ slightly reduces the OA due to excessive feature splitting, $g = 4$ maintains the optimal performance across all datasets. This outcome indicates that $g = 4$ provides a sufficient degree of freedom for the gating mechanism to perform diverse feature modulation without overfragmenting the semantic embeddings.

D. Analysis of Failure Cases

To better understand the limitations of GLIFFM, we examine the confusion matrices and identify failure cases across three benchmark datasets: WHDLD, EuroSAT, and LCZGBA. For each dataset, we select some categories with relatively low classification accuracy and analyze the potential causes of misclassification.

1) WHDLD: As discussed previously, although our GLIFFM achieves competitive performance on the WHDLD dataset and outperforms other baseline models in the school category, the performance gain remains relatively limited. A majority of school scenes are misclassified as urban dwellings. This bottleneck is primarily attributed to the significant interclass similarity between these two categories, which share nearly identical textures, spatial layouts, and spatial contexts. A key discriminative clue for distinguishing the school from urban dwellings is the presence of football fields and the complexity of buildings. However, despite the fact that the football field is visually identifiable in the optical image [see Fig. 12(a)], the U-Net++ module incorrectly merges the small or texturally complex football field with its surrounding areas. Consequently, it is misclassified as an "irregular lawn" in the LC map. This leads to a misleading spatial prior that is indistinguishable from the actual lawns in urban dwellings shown in Fig. 12(b), misleading the model toward the wrong category. In contrast, Fig. 12(c) demonstrates a successful case where the football field exhibits greater spatial dominance and is more accurately delineated in the LC map, thereby providing a robust semantic cue.

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TABLE VIII
PERFORMANCE COMPARISON OF DIFFERENT LC INTEGRATION STRATEGIES ACROSS DIFFERENT DATASETS

Strategy	Fusion Point	LC Semantics	WHDLD(8:2)		WHDLD(5:5)		EuroSAT(5:5)		EuroSAT(2:8)		LCZGBA(5:5)	
			OA (%)	Kappa (%)	OA (%)	Kappa	OA (%)	Kappa (%)	OA (%)	Kappa (%)	OA (%)	Kappa (%)
Baseline	-	-	73.79	69.32	72.79	68.36	93.57	92.84	88.61	87.32	74.27	67.77
Late Fusion	classifier head	global stats	75.30	71.16	73.55	69.15	94.56	93.95	89.46	88.27	76.60	70.74
Ours	**interactive information embedding**	**semantic guidance and interactive features**	**77.12**	**73.34**	**74.48**	**70.31**	**95.43**	**94.91**	**91.61**	**90.66**	**78.66**	**73.32**

TABLE IX
PERFORMANCE COMPARISON OF DIFFERENT GROUP NUMBERS g IN GIFB

Dataset	Metric(%)	$g = 2$	$g = 4$	$g = 8$
WHDLD(8:2)	OA	76.21	**77.12**	76.71
	Kappa	72.33	**73.34**	72.97
WHDLD(5:5)	OA	73.96	**74.48**	74.16
	Kappa	69.68	**70.31**	69.97
EuroSAT(5:5)	OA	95.33	**95.43**	95.35
	Kappa	94.81	**94.91**	94.82
EuroSAT(2:8)	OA	91.43	**91.61**	90.77
	Kappa	90.46	**90.66**	89.73
LCZGBA(5:5)	OA	78.41	**78.66**	78.23
	Kappa	72.99	**73.32**	72.80

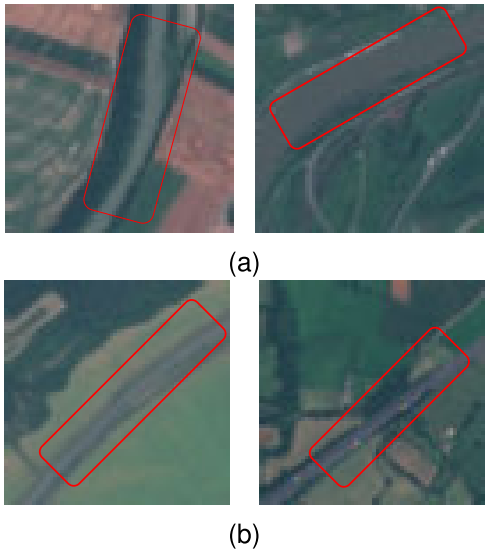


Fig. 13. Visual analysis of classification results on the EuroSAT dataset. (a) Failure case of river. (b) Case of highway.

2) EuroSAT: Another significant classification challenge arises from the river category, which is frequently misidentified as a highway. As illustrated in Fig. 13, there is extreme similarity of the two categories in both spatial morphology and spectral signatures. To be specific, both categories often appear as strip-like structures with similar vegetated surroundings. As shown in the left cases of Fig. 13(a) and (b), certain river segments exhibit a yellowish spectral signature that closely resembles asphalt or concrete roads. Furthermore, when the distinctive spectral signature of water is weakened by turbidity or shadows, the spectral–spatial distinction between river and road diminishes.

3) LCZGBA: The compact high-rise category is rarely misclassified as compact mid-rise, due to its high structure similarity in lower-spatial-resolution images [see Fig. 7(c)] and extreme data scarcity (only 19 training samples). Consequently, the model exhibited a strong bias toward the majority similar class, failing to generalize effectively under structural similarity and severe class imbalance.

In summary, these classification failures arise from semantic degradation in the LC map generation stage, where discriminative structural features are lost or merged into generic land-cover representations. This is compounded by strong interclass similarity and extreme sample scarcity, which together mislead the model's spatial priors and bias predictions toward majority classes under ambiguous conditions.

E. Future Directions

To address the identified bottlenecks, future research will focus on several key areas. First, incorporating multiscale skip-attention mechanisms could mitigate semantic degradation by preserving fine-grained structural cues during the LC map generation stage. Second, the integration of supervised contrastive learning is expected to enlarge the feature-space distance between categories with high interclass similarity, such as rivers and highways. Finally, to overcome the challenge of extreme class imbalance in datasets like LCZGBA, we plan to implement cost-sensitive loss functions and explore generative data augmentation techniques to synthesize high-quality samples for minority classes, thereby enhancing the model's robustness and generalization capabilities in complex RS scenes.

VI. CONCLUSION

In this article, a novel approach for LU classification, i.e., GLIFFM, has been proposed. In addition to the optical information from RS images, GLIFFM learns to leverage deeper, more refined LC primitives. The model learns the intricate relationships between these two data types, fusing their unique characteristics to ultimately classify the LU categories.

First, an LC map is produced using a U-Net++ architecture. It is noteworthy that U-Net++ is exclusively employed in the preprocessing stage to generate LC maps as auxiliary semantic priors, making this process functionally separable from the core GLIFFM architecture. The purpose of this step is to extract the LC primitives relevant to the given LU category. Second, the MFE module is then applied to interactively fuse the optical and LC features. This process is designed to leverage the fact that each LU category is characterized by a specific distribution of different LC types, thereby imparting the LU classification with rich, semantic LC information.

Then, following the feature fusion, the features are processed by a CNN–Transformer to generate a set of global–local

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features with embedded LC information. Directly using this complex feature set for classification, however, can result in errors caused by information redundancy. To overcome this drawback, we introduce the LGFEM and GIFB. The combined use of the LGFEM and GIFB can effectively mitigate this issue and improve classification accuracy. The positive experimental results on the three datasets (i.e., WHDLD, EuroSAT, and LCZGBA) have demonstrated the efficacy of GLIFFM on LU classification tasks.

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